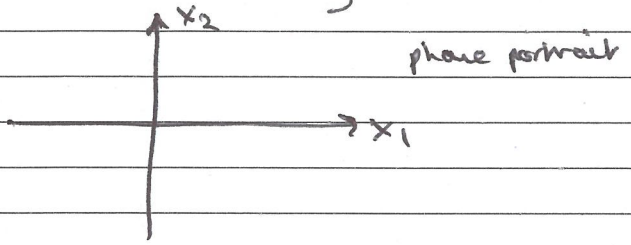
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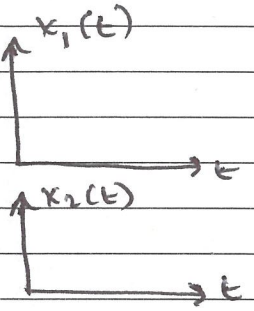
Solving linear systems - review

- Solve $\frac{d\underline{x}}{dt} = \underline{\underline{A}}\,\underline{x}$ for $\underline{x}(t)$ $\qquad \underline{x} = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$
- $|\underline{\underline{A}} - r\underline{\underline{I}}| = 0 \rightarrow$ eigenvalues r_1, r_2
- Solve $A\underline{\xi} = r\underline{\xi}$ for eigenvectors $\underline{\xi}^{(1)}, \underline{\xi}^{(2)}$

General solution

$$\underline{x}(t) = c_1 e^{r_1 t}\underline{\xi}^{(1)} + c_2 e^{r_2 t}\underline{\xi}^{(2)}$$





Recap - Second order ODE

$$ay'' + by' + cy = 0$$

$$y = e^{mt}$$

$$am^2 + bm + c = 0$$

$$y = c_1 e^{m_1 t} + c_2 e^{m_2 t}$$

$m_1 = i\mu$ $(\lambda = 0)$ $\qquad m_2 = -i\mu$

$$y = c_1 e^{i\mu t} + c_2 e^{-i\mu t}$$ — complex sol^{n}

$$e^{i\mu t} = \cos\mu t + i\sin\mu t$$

$$y = c_1(\cos\mu t + i\sin\mu t) + c_2(\cos\mu t - i\sin\mu t)$$
$$= (c_1 + c_2)(\cos\mu t) + i(c_1 - c_2)\sin\mu t$$
$$= A\cos\mu t + B\sin\mu t$$ — real solutions

$A = c_1 + c_2$
$B = i(c_1 - c_2)$

§ 4.4.3 Repeated Roots

$$\underline{x}^{(1)} = \underline{\xi} e^{rt}$$

Try $\underline{x}^{(2)} = \underline{\xi} t e^{rt} + \underline{\eta} e^{rt}$

Example 4.14

$$\underline{\underline{A}} = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}$$

$$|\underline{\underline{A}} - r\underline{\underline{I}}| = 0 \Rightarrow \begin{vmatrix} 1-r & -1 \\ 1 & 3-r \end{vmatrix} = 0 \Rightarrow (1-r)(3-r) + 1 = 0$$
$$3 - 4r + r^2 + 1 = 0$$
$$(r-2)^2 = 0$$

Repeated root $r = 2$.

• $r=2$ $\quad A\underline{\xi} = r\underline{\xi}$ $\quad \begin{pmatrix} 1-2 & -1 \\ 1 & 3-2 \end{pmatrix}\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \underline{0}$

$$\left.\begin{aligned} -\xi_1 - \xi_2 &= 0 \\ \xi_1 + \xi_2 &= 0 \end{aligned}\right\} \quad \text{Take } \underline{\xi}^{(1)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

we have one solution $\underline{x}^{(1)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t}$.

Second solution try $\underline{x}^{(2)} = \underline{\xi} t e^{rt} + \underline{\eta} e^{rt}$

$$\frac{d\underline{x}^{(2)}}{dt} = \underline{\xi} t e^{rt} r + e^{rt}\underline{\xi} + \underline{\eta} e^{rt} r$$

$$A\underline{x}^{(2)} = A\left(\underline{\xi} t e^{rt} + \underline{\eta} e^{rt}\right)$$

Sub. into $\frac{d\underline{x}^{(2)}}{dt} = A\underline{x}^{(2)}$

$$e^{rt}\left(\underline{\xi} t r + \underline{\xi} + \underline{\eta} r\right) = e^{rt}\left(A\underline{\xi} t + \underline{\eta}\right)$$

For this to be valid for all t, we equate coefficients of $t e^{rt}$ and e^{rt} on both sides

$$\underline{\xi} r = A\underline{\xi} \quad (1) \qquad \underline{\xi} + \underline{\eta} r = A\underline{\eta} \quad (2)$$

we have already solved (1) : $\underline{\xi}^{(1)}$ and r_1. To satisfy (2)

(2) $\quad (A - rI)\underline{\eta} = \underline{\xi}$

$$\begin{pmatrix} 1-2 & -1 \\ 1 & 3-2 \end{pmatrix}\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\left.\begin{aligned} -\eta_1 - \eta_2 &= 1 \\ \eta_1 + \eta_2 &= -1 \end{aligned}\right\}$$

if $\eta_1 = k$ then $\eta_2 = -1 - k$ i.e. $\underline{\eta} = \begin{pmatrix} k \\ -1-k \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix} + k\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

and so the solution is

$$\underline{x}^{(2)}(t) = \underline{\xi} t e^{rt} + \underline{\eta} e^{rt}$$

$$= \begin{pmatrix} 1 \\ -1 \end{pmatrix} t e^{2t} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} e^{2t} + k\underbrace{\begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t}}_{\text{multiple of } \underline{x}^{(1)}}$$

$\therefore$ ignore ($k=0$)

$$W[\underline{x}^{(1)}, \underline{x}^{(2)}](t) = \begin{vmatrix} e^{2t} & te^{2t} \\ -e^{2t} & -te^{2t}+e^{2t} \end{vmatrix}$$

$\underline{x}^{(1)} = \underline{\xi} e^{2t}$ $\qquad$ $\underline{x}^{(2)}$

$$= e^{4t}(-t-1+t)$$
$$= -e^{4t} \neq 0 \quad \therefore \ \underline{x}^{(1)} \text{ and } \underline{x}^{(2)} \text{ are L.I}$$

and form a fundamental set.

The general solution is

$$\underline{x}(t) = c_1 \underline{x}^{(1)} + c_2 \underline{x}^{(2)}$$
$$= c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{2t} + c_2 \left[\begin{pmatrix} 1 \\ -1 \end{pmatrix} te^{2t} + \begin{pmatrix} 0 \\ -1 \end{pmatrix} e^{2t} \right] //$$

General case

$$= c_1 \underline{\xi} e^{rt} + c_2 \left[\underline{\xi} te^{rt} + \underline{\eta} e^{rt} \right]$$

$$\underline{x}(t) = e^{rt} \left[(c_1 \underline{\xi} + c_2 \underline{\eta}) + c_2 \underline{\xi} t \right]$$

$$= e^{rt} \underline{y}$$

straight line through the point $(c_1\underline{\xi} + c_2\underline{\eta})$, direction of $\underline{\xi}$

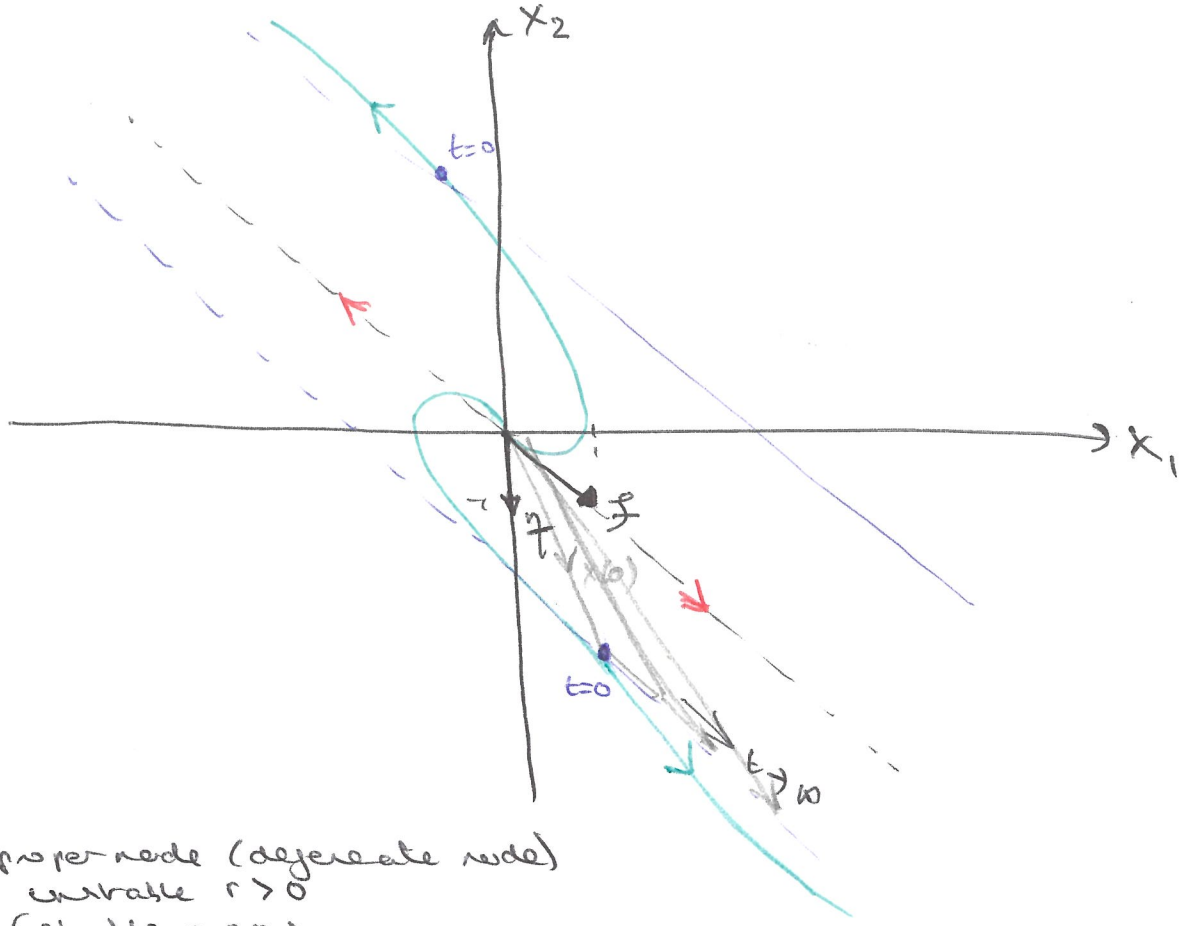
$\underline{y}$ controls the direction

e^{rt} controls $|\underline{x}|$

- Draw $\underline{\xi} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\underline{\eta} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$. Add direction to $\underline{\xi}$ ($c_2 = 0$)
- Identify point $(c_1\underline{\xi} + c_2\underline{\eta})$.
- Particle on the line at $t=0$ moves along in the direction of increasing t
- As $t \to -\infty$ move backwards along $\underline{\xi}$, but $|\underline{x}| \to 0$ as $t \to -\infty$

Plan

- numerical ex
- Wed / formative sheet - Fri
- Summative test Fri (Mock Mon)
- Reading: Stability.



Improper node (degenerate node)
unstable $r > 0$
(stable $r < 0$)